

Security in Cloud Networking

1. In AWS, create a new security group named InstaClone-SG and configure it to allow incoming HTTP (port 80) and HTTPS (port 443) traffic only from your own IP address.

ANS :

1. Log in to the **AWS Management Console**.
2. Open **EC2** → **Security Groups**.
3. Click **Create Security Group**.
4. Enter the following details:
 - **Security Group Name:** InstaClone-SG
 - **Description:** Security group for InstaClone web server.
 - **VPC:** Select your VPC.
5. Add the following **Inbound Rules**:

Type	Protocol	Port	Source
HTTP	TCP	80	My IP
HTTPS	TCP	443	My IP

6. Leave the default outbound rule (Allow All) unless instructed otherwise.
7. Click **Create Security Group**.
Result: Only your current public IP address can access the web server on ports **80** and **443**.

2. Set up a security group for a mock Zomato-style backend server that allows incoming MySQL (port 3306) connections only from a specific IP address and blocks all other inbound traffic.

Hint: Use the 'Custom TCP' rule and specify the allowed IP in CIDR format.

ANS :

1. Create a new Security Group named **Zomato-Backend-SG**.
2. Select the appropriate VPC.
3. Configure the following **Inbound Rule**:

Type	Protocol	Port	Source
MySQL/Aurora (Custom TCP)	TCP	3306	192.168.1.100/32 (Example – replace with the required IP address)

4. Do **not** add any other inbound rules.
5. Click **Create Security Group**.

➤ **Result:** Only the specified IP address can connect to the MySQL database. All other inbound traffic is denied.

3. Modify an existing security group to block all outbound traffic except for DNS (port 53) and HTTPS (port 443) to simulate a strict Paytm wallet server environment.

ANS :

1. Open the Security Group you want to modify.
2. Go to the **Outbound Rules** tab.
3. Delete the default rule (**All Traffic** → **0.0.0.0/0**).
4. Add the following outbound rules:

Type	Protocol	Port	Destination
DNS (UDP)	UDP	53	0.0.0.0/0
DNS (TCP)	TCP	53	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

4. Compare security groups and network ACLs by listing two key differences in how they control traffic for a Flipkart-like cloud application.
Hint: Think about the level (instance vs subnet) and direction (inbound/outbound) each operates at.

ANS :

Feature	Security Groups	Network ACLs
Level of Operation	Applied to individual EC2 instances (instance level).	Applied to entire subnets (subnet level).
Traffic Rules	Stateful – return traffic is automatically allowed.	Stateless – inbound and outbound rules must both be configured explicitly.

- **Explanation**

- For a Flipkart-like cloud application, **Security Groups** protect individual web servers, application servers, and database servers by controlling which traffic can reach each instance. **Network ACLs** provide an additional layer of security at the subnet level, filtering traffic before it reaches any instance within that subnet.